

Yathin Reddy Duvuru

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EDUCATION

New York University, Tandon School of Engineering

Master of Science in Computer Science (GPA - 3.95/4.0)

New York, NY

Sep 2025 - May 2027

SRM Institute of Science and Technology

Bachelor of Technology in Computer Science and Engineering (GPA - 9.49/10.0)

Chennai, India

Sep 2021 - May 2025

TECHNICAL SKILLS

ML: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, MLFlow, SHAP, Captum, Accelerate

Generative AI: LangChain, LangGraph, LlamaIndex, Nano-GraphRAG, Qdrant, ChromaDB, Ollama, Chainlit, AutoResearch, Coding CLIs (Codex, Claude Code, OpenCode)

Languages & Backend: Python, C/C++, SQL, JavaScript, FastAPI, Flask, PostgreSQL, MongoDB

Cloud & DevOps: AWS, Docker, Kubernetes, Terraform, Git, Linux, ArgoCD, Travis CI, Postman

Data Analysis: NumPy, Matplotlib, Seaborn

EXPERIENCE

New York University, Center for Urban Science and Progress

TREx Research Fellow

Brooklyn, NY

June 2026 – Present

- Created a **comparative evaluation pipeline** for a fine-tuned RemoteCLIP model (RoofNet), using **Transformer Explainability, GroundingDINO, and SAM** to perform **model attention analysis** on the hold-out set of 625 satellite images.
- Designed **preprocessing pipelines for external remote sensing datasets**, including class remapping and crop generation, to **benchmark frozen-weight transfer learning performance** of the RoofNet model.
- Migrated the GeoFlood Studio platform from Cesium Ion to a **self-hosted TiTiler/Nginx/GDAL architecture** on NYU on-prem infrastructure, cutting cloud dependencies and **increasing storage capacity 10x**.

New York University RIT, AI Center of Excellence

Full Stack GenAI Developer

Brooklyn, NY

June 2026 – Present

- Engineered a **reusable multimodal embedding layer** with an **RQ-driven durable reindexing pipeline**, enabling **dynamic embedding model selection** for knowledge bases and eliminating the platform's previous single-model embedding constraint.

New York University, Stern School of Business

Graduate Assistant Developer

Brooklyn, NY

Mar 2026 – June 2026

- Architected the migration of the CI Catalyzer platform to a **Multi-Agent LangGraph architecture**, increasing task completion rates by **35%** through autonomous orchestration and fault isolation.
- Integrated **LangSmith** for full-stack observability and evaluation, identifying bottlenecks to **reduce sub-agent task-execution latency by 22%** improving user experience across complex tasks.
- Designed a **high-concurrency, tiered rate-limiting** infrastructure using **AWS API Gateway** and **Lambda**, leveraging **Amazon DAX** clusters to achieve **sub-millisecond lookup times ($\leq 10\text{ms}$)** with DynamoDB fallback.

SRM Institute of Science and Technology

Undergraduate Researcher

Chennai, India

Aug 2024 – May 2025

- Researched **PyTorch-based LSTM** networks for **time-series forecasting** in solar power, and integrated eXplainable AI (XAI) achieving a **40% reduction in training time** by guiding downstream feature selection.
- Optimized model architectures and engineered domain-specific features using **polynomial feature augmentation**, achieving a **10% RMSE reduction** over traditional statistical and regression baselines.
- Developed a **Genetic Algorithm** script for **automated hyperparameter optimization**, reducing RMSE by an **additional 4%** while minimizing manual tuning effort.
- Awarded the **Department Silver Medal** at SRM Research Day 2025.

PROJECTS

Full-Stack RAG Document QA System <i>FastAPI, React, Qdrant, Redis, Celery, PostgreSQL</i>	Jun 2026
- Engineered a hybrid RAG pipeline using Reciprocal Rank Fusion (RRF) to fuse BM25 and semantic vector search, dynamically filtering candidate documents down to the top-k optimal context chunks for SSE answer streaming. - Developed an asynchronous distributed ingestion engine using Celery to handle parallel document processing, achieving a throughput of 100+ pages/minute across text extraction, chunking, and 1,536-dim embedding generation. - Designed a Redis HNSW semantic cache utilizing a two-pass hydration strategy, cutting downstream LLM API costs by serving repetitive queries with a sub-75ms response latency. - Implemented a stateful multi-turn conversation engine with strict per-session document scoping, ensuring isolated user context windows and zero cross-tenant data leakage during vector retrieval.	
Automated Recipe Normalization Engine <i>Triton, Kubernetes, Ansible, MLflow, HF Accelerate</i>	May 2026
- Fine-tuned T5 (Small/Base/Large) variants on a 150K recipe dataset, achieving a 0.70 ROUGE-L score in correcting text errors generated via a custom synthetic data corruption pipeline. - Accelerated multi-GPU training by approx. 2x utilizing FP8 precision over BF16 via the Hugging Face Accelerate library, tracking comprehensive metrics and artifacts seamlessly within MLflow. - Provisioned a 5-node Kubernetes cluster on Chameleon Cloud via Ansible, reducing infrastructure setup time by 90% (to under 12 minutes) through end-to-end configuration playbooks. - Deployed models to production via Triton Inference Server to guarantee sub-3s inference latency, monitoring cluster health and performance with Prometheus and Grafana.	
GraphRAG-Based AI Tutor <i>Nano-GraphRAG, Qdrant, Chainlit, Ollama, Docker</i>	Dec 2025
- Implemented an end-to-end Generative AI pipeline for scraping, chunking, and indexing high-volume datasets using as feature embeddings in a Qdrant vector store. - Built a GraphRAG system integrating vector similarity search with knowledge graph traversal (Nano-GraphRAG), enabling multi-hop reasoning and increasing answer relevance by 30% over baseline. - Developed an automated graph construction workflow using Qwen3-32B, reducing manual schema design effort by 80% while processing 255+ documents for downstream retrieval.	
Basketball Video Analytics Pipeline <i>Python, PyTorch, OpenCV, Roboflow, SAM2</i>	Nov 2025
- Engineered a Computer Vision pipeline processing 100K+ frames, utilizing PyTorch and OpenCV for player and ball detection with Roboflow models. - Improved pipeline performance by replacing SAM2-based segmentation with ByteTrack tracking, achieving a 27x speedup (for a 20-minute clip) to resolve compute bottlenecks. - Added a local RAG-based QA system over 10K+ tokens of game commentary using Ollama and Qdrant for context-aware gameplay insights.	

PUBLICATIONS

Yathin Reddy D., Seshank M., Saranya P. <i>Photovoltaic Cell Power Forecasting Using LSTM with XAI Integration</i> . Springer LNNS, 2024. https://doi.org/10.1007/978-981-96-7514-2_22	
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CERTIFICATIONS

Stanford University: Machine Learning Specialization	<i>Sept. 2024</i>
AWS Academy Graduate: Cloud Foundations	<i>June 2023</i>
AWS Academy Graduate: AWS Academy Machine Learning Foundations	<i>April 2023</i>
Palo Alto Networks: Cybersecurity Virtual Internship	<i>March 2023</i>